Complex fourth-root approximations: analytic limits, positive source calculus and clock channels

Project Stardust

Reviewed editorial draft, October 29, 2026

Abstract

Uniform complex fourth-root steps yield genuine positive probability laws whose holomorphic exponential transforms approach the biharmonic heat symbol. Their absolute second moments diverge, and exact fourth moments give a uniform positive mass outside each compact set, eventually along every refinement. Thus the transform limit is compatible with the absence of any weak probability-law limit. Poissonization preserves both features. Normalized complex exponential tilts have mass one and convergent holomorphic transforms, but their total variations diverge. We exhibit a single bounded continuous test, independent of the approximation index, whose integral equals the total variation exactly. The paper reconstructs verified fourth-order results with elementary proofs and explicit constants. The analytic-versus-probabilistic distinction is established in prior work; this is a formalization companion, and no priority claim is made. The actual mark source separately has a positive Hermitian three-channel predictable calculus, full original-source L^2 reconstruction and a distinct Poisson clock channel. Complete proofs retain the difference between this source calculus and its analytic scaling target. A compact-frequency domain supports genuine integrable transitions, derivative limits and finite-process backward martingales.

1 Scope and main conclusions

Let ε_j, δ_j be mutually independent symmetric signs and set

$$\zeta_j = \varepsilon_j + i\delta_j, \qquad S_{n,a} = a\sum_{j=1}^{n} \zeta_j, \qquad a_n(T) = (6T/n)^{1/4} \quad (n \geq 1).$$

The four values of ζ_j are the fourth roots of -4. Write μ_n^T for the law of $S_{n,a_n(T)}$ on $\mathbb{C}$ with its usual Euclidean topology. For $r \geq 0$, let N_r be Poisson with mean r, independent of the signs, and let $\mu_{r,a}$ be the law of

$$C_{r,a} = a\sum_{j=1}^{N_r} \zeta_j.$$

All these laws are ordinary positive probability measures.

Theorem 1.1. *For $T > 0$ and each real η,*

$$\int e^{i\eta z}\, d\mu_n^T(z) \longrightarrow e^{-T\eta^4}, \qquad \int e^{i\eta z}\, d\mu_{n,a_n(T)}(z) \longrightarrow e^{-T\eta^4}.$$

Nevertheless, for every compact $K \subset \mathbb{C}$, eventually in n,

$$\mu_n^T(K^c) \geq \tfrac{1}{8}, \qquad \mu_{n,a_n(T)}(K^c) \geq \tfrac{1}{12}.$$

Neither family is tight. Neither has a weak probability-law limit along any reindexing $n = \varphi(k)$ with $\varphi(k) \to \infty$.

The transforms in this statement test $e^{i\eta z}$, which is unbounded on $\mathbb{C}$ when $\eta \neq 0$. They are not the characteristic functions $e^{i(u \operatorname{Re} z + v \operatorname{Im} z)}$ used in weak convergence on $\mathbb{R}^2$. This domain distinction is essential.

For the tilted result use the second, rate-scaled normalization $b_n = a_n(1)$, $r_n = nT$. Define real functions

$$F(x) = \cos x \cosh x, \qquad \psi_a(\xi) = F(a\xi) - 1,$$

and the finite complex measure

$$d\nu_n(z) = e^{-r_n \psi_{b_n}(\xi)} e^{i\xi z}\, d\mu_{r_n, b_n}(z). \tag{1}$$

Here and below total variation means the mass $|\nu|(\mathbb{C})$, with no factor $1/2$.

Theorem 1.2. *Fix $T > 0$ and $\xi \in \mathbb{R} \setminus \{0\}$. Each ν_n has mass one, and for every real η,*

$$\int e^{i\eta z}\, d\nu_n(z) \longrightarrow e^{-T((\eta+\xi)^4 - \xi^4)}. \tag{2}$$

The fixed function $f_\xi(z) = e^{-i\xi \operatorname{Re} z}$ is continuous and has modulus one, and satisfies

$$\int f_\xi\, d\nu_n = |\nu_n|(\mathbb{C}) = \exp(r_n \cosh(b_n\xi)(1 - \cos(b_n\xi))) \longrightarrow +\infty. \tag{3}$$

Consequently these complex measures do not converge against all bounded continuous tests to any finite complex measure, even along a divergent reindexing.

Relationship with prior work. Bonaccorsi and Mazzucchi [1] construct higher-order analytic limits from complex root walks. Their Theorem 2 treats exponential transforms; their Remark 1 discusses escape from bounded regions, and their discussion before Theorem 7 distinguishes the analytic limit from an ordinary limit process. Our rotated fourth roots and normalization are a fourth-order instance of that construction. We do not assert that their analytic framework supplies an ordinary weak limit. The contribution of this companion is a complete reconstruction of the frozen formal results: finite moment identities, explicit escape bounds, Poissonized statements, and a fixed bounded test for actual complex coordinate measures. These are meaningful precise distinctions, but their independent novelty has not been established. The later analytic Itô framework [2] is also separate from weak convergence of the present probability laws.

2 Transforms and integrability

Lemma 2.1. *For real a, η,*

$$\mathbb{E} e^{i\eta a \zeta_1} = F(a\eta), \tag{4}$$

$$\mathbb{E}\left|e^{i\eta a \zeta_1}\right| = \cosh(a\eta). \tag{5}$$

Consequently

$$\mathbb{E} e^{i\eta S_{n,a}} = F(a\eta)^n, \tag{6}$$

$$\mathbb{E} e^{i\eta C_{r,a}} = \exp(r(F(a\eta) - 1)), \tag{7}$$

$$\mathbb{E}\left|e^{i\eta C_{r,a}}\right| = \exp(r(\cosh(a\eta) - 1)). \tag{8}$$

All integrals in these formulas are absolutely integrable.

Proof. Independence of the two signs gives $\mathbb{E}e^{ix(\varepsilon+i\delta)} = (\mathbb{E}e^{ix\varepsilon})(\mathbb{E}e^{-x\delta}) = \cos x \cosh x$. The modulus is $e^{-x\delta}$, which gives (5). Independence over j gives products for the walk, including $\mathbb{E}|e^{i\eta S_{n,a}}| = \cosh(a\eta)^n$. Conditioning on $N_r = m$ and summing the Poisson series gives

$$e^{-r}\sum_{m=0}^{\infty}\frac{r^m}{m!}F(a\eta)^m = e^{r(F(a\eta)-1)}.$$

Absolute convergence follows either from the same series with $|F(a\eta)|$ or from the nonnegative modulus series $e^{-r}\sum_m r^m\cosh(a\eta)^m/m!$. This also proves (8) and justifies conditioning for the complex integrals. □

Proposition 2.2. *For $T \geq 0$ and real η,*

$$\begin{aligned} F(a_n(T)\eta)^n &\longrightarrow e^{-T\eta^4}, \\ e^{n(F(a_n(T)\eta)-1)} &\longrightarrow e^{-T\eta^4}, \\ e^{nT(F(b_n\eta)-1)} &\longrightarrow e^{-T\eta^4}. \end{aligned}$$

Proof. Multiplication of the convergent cosine and hyperbolic cosine series gives $F(x) = 1-x^4/6+O(x^8)$ as $x \to 0$; the degree-two and degree-six terms cancel. Thus

$$n(F(a_n(T)\eta) - 1) = -T\eta^4 + O(n^{-1}).$$

For the walk, $F(a_n(T)\eta) > 0$ eventually, and $\log(1+y) = y+O(y^2)$ shows that $n\log F(a_n(T)\eta) \to -T\eta^4$. Exponentiation proves the first two limits. Replacing $a_n(T)$ by b_n gives $n(F(b_n\eta)-1) \to -\eta^4$; multiplication by T proves the third. For $T = 0$, the first two sequences are identically one, as is the third. □

These statements establish only the stated exponential-test limits. Passing a general entire function through the limit needs a separate integrability and domination argument. The compact-frequency domain below supplies a justified restricted extension.

3 Exact positive moments and compact escape

Lemma 3.1. *For $n \geq 0$, real a, and $r \geq 0$,*

$$\mathbb{E}|S_{n,a}|^2 = 2na^2, \qquad \mathbb{E}|S_{n,a}|^4 = (8n^2-4n)a^4, \tag{9}$$

$$\mathbb{E}|C_{r,a}|^2 = 2ra^2, \qquad \mathbb{E}|C_{r,a}|^4 = (8r^2+4r)a^4. \tag{10}$$

Proof. Let $R_n = \sum_{j=1}^n \varepsilon_j$ and $I_n = \sum_{j=1}^n \delta_j$. These variables are independent, with $\mathbb{E}R_n^2 = \mathbb{E}I_n^2 = n$. In the fourth power of R_n, only four equal indices and two equal pairs survive sign averaging. Therefore

$$\mathbb{E}R_n^4 = n + 6\binom{n}{2} = 3n^2 - 2n.$$

Since $|S_{n,a}|^2 = a^2(R_n^2 + I_n^2)$, expansion gives

$$\mathbb{E}|S_{n,a}|^4 = a^4\big(2(3n^2-2n) + 2n^2\big),$$

proving (9). The defining Poisson series gives $\mathbb{E}N_r = r$ and $\mathbb{E}[N_r(N_r-1)] = r^2$ by shifting the summation index. Hence $\mathbb{E}N_r^2 = r^2+r$. Conditioning (9) on N_r proves (10). All these nonnegative moments are finite by the same Poisson moment sums. □

The positive second moment in this lemma must be distinguished from $\mathbb{E}S_{n,a}^2 = 0$. The latter is a holomorphic cancellation and is not a variance in the complex Hilbert-space sense.

Lemma 3.2. *If $X \geq 0$, $0 < m = \mathbb{E}X < \infty$, and $\mathbb{E}X^2 \leq Am^2$, then for $0 \leq b \leq m/2$,*

$$\mathbb{P}(X > b) \geq \frac{1}{4A}.$$

Proof. Split the expectation at b and apply Cauchy–Schwarz:

$$m \leq b + \mathbb{E}[X\mathbf{1}_{\{X>b\}}] \leq b + (\mathbb{E}X^2)^{1/2}\mathbb{P}(X > b)^{1/2}.$$

Thus $m/2 \leq \sqrt{A}m\mathbb{P}(X > b)^{1/2}$. Divide by m and square. □

Proof of Theorem 1.1. The transforms follow from Lemma 2.1 and Proposition 2.2. At $a = a_n(T)$, the common second moment of the walk and the rate-n compound law is

$$m_n = 2\sqrt{6Tn} \longrightarrow \infty.$$

For the walk the fourth moment is $(48n-24)T \leq 2m_n^2$; for the compound law it is $(48n+24)T \leq 3m_n^2$, since $n \geq 1$. Given compact K, choose $R > 0$ with $K \subset \{|z| \leq R\}$. Eventually $m_n \geq 2R^2$. Apply Lemma 3.2 to $X = |S|^2$ or $|C|^2$, with $b = R^2$ and $A = 2$ or 3 respectively. The event $\{X > R^2\}$ lies in K^c, giving the two constants.

This contradicts tightness with any error smaller than $1/12$. The same eventual inequalities hold after any $\varphi(k) \to \infty$. To rule out a weak limit directly, let ρ be a proposed probability limit. The disks $\{|z| \leq R\}$ increase to $\mathbb{C}$, so choose R with $\rho(\{|z| \leq R\}) > 1 - 1/48$. Let χ be a continuous function equal to one on that disk, zero outside the disk of radius $R+1$, and between zero and one. Then $\int \chi \, d\rho > 1 - 1/48$, while every sufficiently large approximating law has $\int \chi \leq 1 - 1/12$. This contradicts convergence against the fixed bounded continuous χ, and proves the assertion for both families and every divergent reindexing. □

The constants are elementary uniform lower bounds, not claimed optimal. In particular this proof does not claim that the escaping mass tends to one; the verified conclusion is already enough to exclude weak limits.

4 Path-space consequences with an explicit topology

For $n \geq 1$, join the successive partial sums $a\sum_{j=1}^{k}\zeta_j$ linearly at times k/n to obtain a continuous path $\Gamma_{n,a} : [0,1] \to \mathbb{C}$. Give $C([0,1];\mathbb{C})$ the uniform norm. Its endpoint is $S_{n,a}$.

Corollary 4.1. *For $T > 0$, the laws of $\Gamma_{n,a_n(T)}$ are not tight in the uniform path topology. More generally, any family of probability measures on a topological path space admitting a continuous coordinate map with coordinate laws μ_n^T or $\mu_{n,a_n(T)}$ is not tight.*

Proof. A continuous image of a compact set is compact. If a family of path laws were tight, for every $\epsilon > 0$ one could choose a compact path set carrying mass at least $1-\epsilon$ under every member. Its coordinate image would be a compact subset of $\mathbb{C}$ carrying at least that mass under every coordinate law. This contradicts Theorem 1.1. Endpoint evaluation is continuous in the uniform norm, which proves the first assertion. □

This implication specifies its topology. The formal Poisson trajectory corollary uses a continuous coordinate on a countable product space. It is not silently a statement about every Skorokhod topology: evaluation at a jump time requires its own continuity hypotheses there. No path-law limit is constructed from the holomorphic transform alone.

5 Actual complex tilts and the fixed test

We prove Theorem 1.2 through formulas valid at every finite parameter. Fix $a, \xi \in \mathbb{R}$, $r \geq 0$, and put

$$h_{r,a,\xi}(z) = e^{-r\psi_a(\xi)} e^{i\xi z}, \qquad d\nu_{r,a,\xi} = h_{r,a,\xi}\, d\mu_{r,a}.$$

The normalizing factor is a positive real number because $\psi_a(\xi)$ is real, although $\psi_a(\xi)$ need not be nonnegative.

Proposition 5.1. *The density is absolutely integrable, $\nu_{r,a,\xi}(\mathbb{C}) = 1$, and*

$$\int e^{i\eta z}\, d\nu_{r,a,\xi}(z) = \exp\big(r(\psi_a(\eta+\xi) - \psi_a(\xi))\big), \tag{11}$$

$$|\nu_{r,a,\xi}|(\mathbb{C}) = \exp\big(r\cosh(a\xi)(1-\cos(a\xi))\big). \tag{12}$$

The exponential test in (11) *is integrable for the total variation measure. Furthermore*

$$f_\xi(z) h_{r,a,\xi}(z) = |h_{r,a,\xi}(z)| \qquad (z \in \mathbb{C}), \tag{13}$$

so $\int f_\xi\, d\nu_{r,a,\xi} = |\nu_{r,a,\xi}|(\mathbb{C})$.

Proof. Formula (8) gives density integrability. Multiplying the density by $e^{i\eta z}$ leaves the same positive normalizer times $e^{i(\eta+\xi)z}$, whose modulus is integrable by that formula again. The compound transform (7) now proves (11); taking $\eta = 0$ proves mass one.

For completeness, the variation of a finite complex density h relative to a positive measure μ is $|h|\mu$. The upper bound follows from $|\int_A h\, d\mu| \leq \int_A |h|\, d\mu$ on every measurable partition. For the reverse bound, approximate the measurable phase $h/|h|$ on $\{h \neq 0\}$ uniformly by finitely many phases on a partition of the unit circle into arcs of arbitrarily small diameter. On each resulting measurable piece the integral differs from its constant-phase integral by at most that diameter times $\int |h|$. Summing and letting the diameter vanish proves equality for every measurable set. In particular,

$$\begin{aligned} |\nu_{r,a,\xi}|(\mathbb{C}) &= e^{-r(F(a\xi)-1)} \mathbb{E}|e^{i\xi C_{r,a}}| \\ &= e^{-r(F(a\xi)-1)+r(\cosh(a\xi)-1)} \\ &= e^{r\cosh(a\xi)(1-\cos(a\xi))}. \end{aligned}$$

Finally, writing $z = x + iy$, one has $e^{-i\xi x} e^{i\xi z} = e^{-\xi y} = |e^{i\xi z}|$. The normalizer is positive real, so the same identity holds for the density. This is (13). Integrating it proves the last claim. The test depends on ξ alone, not on a or r. □

Proof of Theorem 1.2. Apply Proposition 5.1 with $a = b_n$, $r = nT$. Proposition 2.2 gives $nT\psi_{b_n}(u) \to -Tu^4$ for every real u. Subtraction at $u = \eta + \xi$ and $u = \xi$ followed by exponentiation proves (2).

To prove divergence using a quantitative lower bound, observe that $|b_n\xi| \to 0$. For $|x| \leq \pi$, concavity of sine on $[0, \pi/2]$ gives $\sin(|x|/2) \geq |x|/\pi$. Consequently

$$1 - \cos x = 2\sin^2(x/2) \geq 2x^2/\pi^2.$$

Since $\cosh x \geq 1$, eventually

$$nT\cosh(b_n\xi)(1-\cos(b_n\xi)) \geq \frac{2T\xi^2}{\pi^2}\, nb_n^2 = \frac{2T\xi^2\sqrt{6}}{\pi^2}\sqrt{n} \longrightarrow \infty. \tag{14}$$

The coefficient is positive by $T > 0$ and $\xi \neq 0$. Equation (12) and the phase cancellation give (3). A finite complex measure has a finite integral against the modulus-one function f_ξ. Divergence of this one integral excludes convergence against all bounded continuous tests. The same lower bound holds after every divergent reindexing. □

The exclusions $T > 0$, $\xi \neq 0$ are indispensable. If $T = 0$ the law is δ_0. If $\xi = 0$ the density is one and the variation is one, although positive-law compact escape can still occur at positive T. Nor is the fixed-test conclusion merely a restatement that some sequence of test functions detects growing variation: a single explicitly given test works for the entire sequence.

6 Cyclic mark channels and positive Hermitian energy

The probability sources behind the preceding approximations admit an exact calculus before any scaling limit. Its positive Hermitian energy must be distinguished from its vanishing holomorphic moments and from the signed higher-heat target. We first record the finite mechanism.

Let K be uniform on $\mathbb{Z}/m\mathbb{Z}$, $m \geq 1$, and put $\chi_j(k) = e^{2\pi ijk/m}$. The elementary character sum is

$$\frac{1}{m}\sum_{k=0}^{m-1} \chi_j(k) = \mathbf{1}_{j=0}.$$

For $j \neq 0$, multiply the geometric sum by $1 - e^{2\pi ij/m}$ to prove it is zero; at $j = 0$ it is one after normalization. Consequently $\mathbb{E}[\chi_j(K)\overline{\chi_l(K)}] = \mathbf{1}_{j=l}$. For any function g on the finite mark space, define

$$\widehat{g}_j = \frac{1}{m}\sum_k g(k)\overline{\chi_j(k)}.$$

Orthogonality and the dimension m give

$$g(k) - \mathbb{E}g(K) = \sum_{j\neq 0} \widehat{g}_j \chi_j(k), \qquad \mathbb{E}|g(K) - \mathbb{E}g(K)|^2 = \sum_{j\neq 0} |\widehat{g}_j|^2. \tag{15}$$

Here j ranges over the residue classes; the case $m = 1$ has empty centered sum. Fourier sign conventions are fixed by the displayed conjugate; the source uses its matching discrete Fourier transform. If $\zeta = c\chi_1(K)$, then

$$\mathbb{E}\zeta^r = \mathbf{1}_{m|r} c^r, \qquad \mathbb{E}|\zeta|^2 = |c|^2.$$

The first identity is the character sum applied to r; the second uses the pointwise modulus. Thus holomorphic cancellation is not a zero variance in the Hilbert sense.

For the four marks used in this paper, $\xi = a(1+i)i^K$ in their cyclic order. Assume $a \in \mathbb{R}\setminus\{0\}$, and set

$$w_r = (2a^2)^r \quad (1 \leq r \leq 3), \qquad q = -4a^4.$$

Then $\xi^4 = q$ pointwise, $\mathbb{E}\xi^r = 0$ for $1 \leq r \leq 3$, and

$$\mathbb{E}[\xi^r \overline{\xi^s}] = \mathbf{1}_{r=s} w_r. \tag{16}$$

Indeed $\xi^r = a^r(1+i)^r\chi_r(K)$ and its squared modulus is w_r. These are all three nonconstant characters with nonzero normalizations. For every g on the four marks, put

$$c_r(g) = \frac{\mathbb{E}[g\overline{\xi^r}]}{w_r} = \frac{\mathbb{E}[g\xi^{4-r}]}{q}. \tag{17}$$

The equality follows pointwise from $q\overline{\xi^r} = w_r\xi^{4-r}$, using the constant modulus and $\xi^4 = q$. Thus (15) becomes

$$g - \mathbb{E}g = \sum_{r=1}^{3} c_r(g)\xi^r, \qquad \mathbb{E}\left|\sum_{r=1}^{3} u_r\xi^r\right|^2 = \sum_{r=1}^{3} w_r|u_r|^2. \tag{18}$$

Pairing with each conjugate power proves uniqueness. A single first channel cannot represent the centered square ξ^2: if $\xi^2 = b\xi$, multiply by ξ^2 and integrate to obtain $q = b\mathbb{E}\xi^3 = 0$, a contradiction. This obstruction is on a genuine four-point probability space.

7 Predictable sums and finite reconstruction on the mark source

Work on the actual infinite product $\{0,1,2,3\}^{\mathbb{N}}$ with uniform independent coordinates K_n, and write $\xi_n = a(1+i)i^{K_n}$. Let $\mathcal{H}_n = \sigma(K_0,\dots,K_{n-1})$. A predictable coefficient $H_{r,n}$ is a complex function of this prefix. Every such function has all finite moments at each fixed n, because the prefix space is finite; no uniform bound in n is assumed. Define

$$S_N(H) = \sum_{n=0}^{N-1}\sum_{r=1}^{3} H_{r,n}\xi_n^r, \qquad e_n(H) = \sum_{r=1}^{3} w_r|H_{r,n}|^2.$$

Fresh-mark independence and (16) give the actual conditional identities

$$\mathbb{E}[\Delta S_n \mid \mathcal{H}_n] = 0, \tag{19}$$

$$\mathbb{E}[|C + \Delta S_n|^2 \mid \mathcal{H}_n] = |C|^2 + e_n(H) \tag{20}$$

for every complex prefix-measurable C. In the second identity, expand the square into its Hermitian cross terms; the linear terms vanish and the quadratic terms use (16). In particular S_N is a martingale and

$$\|S_N(H)\|_2^2 = \sum_{n<N} \mathbb{E}e_n(H). \tag{21}$$

Iteration of (20) with $C = S_n$ proves this formula; it is not an isometry for the unconjugated square S_N^2.

Theorem 7.1. *For every function F of the first N marks, there are predictable coefficients with*

$$F - \mathbb{E}F = S_N(H).$$

They are unique almost surely for every channel and time $n < N$. More generally, every integrable martingale in this prefix filtration has a coherent three-channel representation of all its increments.

Proof. For the terminal value, set $M_n = \mathbb{E}[F \mid \mathcal{H}_n]$, $n \le N$. Write $M_{n+1} = f_{n+1}(K_0,\dots,K_n)$ and, for a fixed prefix p, apply (18) to $g_p(k) = f_{n+1}(p,k)$. The mean of g_p is $M_n(p)$ by conditioning on the fresh uniform mark. Thus $M_{n+1} - M_n = \sum_r c_r(g_p)\xi_n^r$. These coefficients depend only on the prefix. Summing telescopes to $M_N - M_0 = F - \mathbb{E}F$. For a general martingale use its prefix functions at each successive time in the same construction. This gives one array for all finite horizons, rather than a separately chosen array at each horizon.

If two arrays have the same terminal sum, subtract them and apply (21). Its nonnegative summands have sum zero. For every $n < N$, each positive w_r therefore multiplies a coefficient of zero L^2 norm. This is the stated almost sure uniqueness. It makes no assertion about coefficients beyond the specified terminal horizon. □

The same conditional argument with the $m-1$ nonzero characters gives finite-horizon cyclic predictable integration and its unit-weight isometry for any $m \geq 1$. The source formally supplies this general finite-energy calculation; the completed operator constructed next is mapped specifically to the actual four-mark source.

8 Infinite limits and the exact completed integration operator

A coefficient array has finite energy if

$$\mathcal{E}(H) = \sum_{n=0}^{\infty} \mathbb{E} e_n(H) < \infty. \tag{22}$$

Cross-time martingale orthogonality gives, for $M \geq N$,

$$\|S_M(H) - S_N(H)\|_2^2 = \sum_{n=N}^{M-1} \mathbb{E} e_n(H).$$

Consequently the sums are Cauchy in the actual complex L^2 space. Let JH denote their limit. Passing to norms, means and bounded conditional projections gives

$$\|JH\|_2^2 = \mathcal{E}(H), \quad \mathbb{E}JH = 0, \quad \mathbb{E}[JH \mid \mathcal{H}_N] = S_N(H). \tag{23}$$

Linearity follows by taking limits of the finite sums. Conversely, L^2 convergence of those sums bounds their norms; the nonnegative energy partial sums in (21) are then bounded, so (22) holds. This supplies both directions of the finite-energy convergence criterion.

We give the needed prefix-density argument explicitly. The collection of sets approximable in probability by finite-coordinate cylinder sets is a sigma algebra: it is closed under complements; finite unions follow by the union bound; for a countable union first truncate the union using continuity of a finite measure and then approximate the finite union. It contains all cylinders and hence the product sigma algebra. Indicators of finite-measure sets can therefore be approximated in L^p, for every finite p, by cylinder indicators. Truncation and simple-function approximation extend this density to L^1 and L^2 functions, including measurable representatives modulo null sets. Conditional expectation is a contraction in either norm. If G is a cylinder function measurable at time m, then $\mathbb{E}[G \mid \mathcal{H}_N] = G$ for $N \geq m$. Thus

$$\|\mathbb{E}[F \mid \mathcal{H}_N] - F\|_p \leq 2\|F - G\|_p \longrightarrow 0 \quad (p = 1 \text{ or } 2)$$

by first choosing a good cylinder approximation and then taking $N \geq m$.

Theorem 8.1. *Every centered $F \in L^2$ on the actual infinite four-mark product has a coherent predictable array H with*

$$S_N(H) = \mathbb{E}[F \mid \mathcal{H}_N], \qquad S_N(H) \longrightarrow F \text{ in } L^2, \qquad \mathcal{E}(H) = \|F\|_2^2.$$

The integration range is exactly the closed centered L^2 subspace. For general $F \in L^1$, the coherent representation of $\mathbb{E}[F \mid \mathcal{H}_N] - \mathbb{E}F$ converges in L^1 and almost surely, without an assertion of finite square energy.

Proof. Apply the martingale construction of Theorem 7.1 to $M_N = \mathbb{E}[F \mid \mathcal{H}_N]$ for every N with its single coherent array. Prefix density proves the claimed norm convergence. In the L^2 case, (21) and norm continuity identify the sum of energies with $\|F\|_2^2$. Conversely every finite-energy integral is centered by (23). Centered functions form the kernel of the continuous mean functional, hence a closed subspace.

Here is an elementary justification of the almost sure conclusion in the L^1 case. For a complex integrable V, conditional Jensen makes $|\mathbb{E}[V \mid \mathcal{H}_n]|$ a nonnegative submartingale, and the finite maximal inequality gives

$$P\{\sup_n |\mathbb{E}[V \mid \mathcal{H}_n]| > \lambda\} \le \lambda^{-1}\mathbb{E}|V|.$$

For a finite maximum, stop at the first crossing and use conditional Jensen against the final conditional value to bound λ times the crossing probability by its final mean. Increasing the finite horizon gives the displayed inequality. Choose cylinders G_j with $\mathbb{E}|F - G_j| \le 2^{-3j}$, and choose m_j so G_j is measurable at $\mathcal{H}_{m_j}$. For $n \ge m_j$, the difference $\mathbb{E}[F \mid \mathcal{H}_n] - F$ is bounded in modulus by $|\mathbb{E}[F - G_j \mid \mathcal{H}_n]| + |F - G_j|$. The maximal inequality and Markov's inequality bound the probability that its supremum exceeds 2^{1-j} by 2^{1-2j}. These probabilities are summable. The union bound on their tails, or the first Borel–Cantelli lemma, proves that eventually each supremum is small. This proves almost sure convergence. Subtracting the constant mean and using the coherent identity transfers it to the stochastic sums. □

For precision, the formal Hilbert integrand space is the vector space of finite-energy coefficient arrays modulo the kernel of J, with norm induced by its realization in the actual integration range. Equation (23) identifies this norm with the printed energy. The range theorem identifies it isometrically with centered L^2, so this quotient is complete; no completion is postulated to make surjectivity true. Write $\mathcal{I}$ for that Hilbert space and still write $J : \mathcal{I} \to L^2$ for its isometric inclusion onto the centered range. Its bounded Hilbert adjoint $D = J^*$ satisfies

$$DJ = I_{\mathcal{I}}, \qquad JDF = F - \mathbb{E}F. \tag{24}$$

Indeed $\langle Ju, Jv\rangle = \langle u, v\rangle$ gives $J^*J = I$; JJ^* is the orthogonal projection onto the closed range, which is exactly centering. Thus D is a bounded predictable reconstruction operator. The notation does not identify it with an unbounded anticipating Malliavin gradient on a different integrand space.

9 Poissonizing the source does not remove the clock channel

Now use an independent sequence of unit exponential holding times $E_1, E_2, \ldots$, arrival times $\tau_n = E_1 + \cdots + E_n$, and $N_t = \max\{n : \tau_n \le t\}$. Positivity holds simultaneously almost surely. Moreover the events $E_j \ge 1$ are independent with probability e^{-1}. For each fixed index k, the probability that none occur after k is the limit of $(1 - e^{-1})^l$, namely zero. A countable union shows infinitely many such events occur, so $\tau_n \to \infty$. Thus N_t is finite at every finite t on one common event, and the marked path

$$X_t = \sum_{j=0}^{N_t - 1} \xi_j$$

is an actual right-continuous finite-jump path, with its pre-jump positions and jumps given by the original partial sums and marks.

The exponential renewal calculation gives $P(N_t = n) = e^{-t}t^n/n!$. To see this directly, convolution gives the arrival density $e^{-u}u^{n-1}/(n-1)!$ for $n \geq 1$; integrating it on $0 < u < t$ against the survival $e^{-(t-u)}$ of the next holding time gives the formula. For $n = 0$ it is just exponential survival. Memorylessness $P(E > s + t \mid E > s) = e^{-t}$, together with independence of unused holdings and marks, supplies stationary independent marked future increments at deterministic wall-clock times. Here is the source-level conditioning argument. On $\{N_s = n\}$, condition first on completed holdings $E_1, \ldots, E_n$ with $\tau_n \leq s$ and the first n marks. The only condition on the next holding is $E_{n+1} > u = s - \tau_n$. In its density integral, the substitution $E_{n+1} = u + r$ gives $e^{-u-r}dr = e^{-u}e^{-r}dr$. The factor e^{-u} is precisely the survival weight for this prefix; the residual r therefore has the unit exponential law independently of the prefix. Unused holdings and marks retain their independent product laws. Testing against bounded functions of the prefix and fresh source, and summing over n, proves factorization on this partition. Every coordinate of the observed past is a measurable function of the count and completed prefixes. A monotone-class extension from their cylinder tests therefore proves independence from the entire observed past. Reconstructing the future from the residual holding and unused source gives the claimed increment laws. In particular $N_t - t$ is a genuine count-filtration martingale, since a future count increment has independent Poisson mean $t - s$. This is a separate holding/count source from the mark-only product.

For bounded continuous $f : \mathbb{C} \to \mathbb{C}$, let

$$Pf(z) = \mathbb{E}f(z + \xi), \qquad Af = Pf - f, \qquad C_r f(z) = c_r(k \mapsto f(z + \xi(k))).$$

These are bounded operators in the uniform norm: $\|P\| \leq 1$, $\|A\| \leq 2$ and $\|C_r\| \leq w_r^{-1/2}$. The last bound follows from (17) and the constant modulus of ξ. Continuity follows from finite sums of translates. Formula (18) gives the exact jump identity

$$f(z + \xi) - f(z) = Af(z) + \sum_{r=1}^{3} C_r f(z)\xi^r. \tag{25}$$

It holds for every mark, without a Taylor remainder or analyticity hypothesis on f.

Theorem 9.1. *With $X_{\tau_{j+1}-}$ denoting the actual pre-jump value, define*

$$K_t(f) = \sum_{j<N_t} Af(X_{\tau_{j+1}-}) - \int_0^t Af(X_s)\,ds,$$
$$L_{r,t}(f) = \sum_{j<N_t} C_r f(X_{\tau_{j+1}-})\xi_j^r.$$

Then the exact actual path formula is

$$f(X_t) - f(0) - \int_0^t Af(X_s)\,ds = K_t(f) + \sum_{r=1}^{3} L_{r,t}(f).$$

The combined process is a martingale in the actual observed natural filtration of X. The centered mark span alone cannot represent the unit jump of the count process for every mark.

Proof. Summing (25) at the finitely many jumps telescopes the left observable. Nonexplosion and the exact partial-sum path identify each pre-jump value, and subtracting the displayed time integral gives the identity.

For the martingale assertion, the actual transition semigroup is

$$T_t = e^{-t} \sum_{n=0}^{\infty} \frac{t^n}{n!} P^n = e^{tA}.$$

The series converges in bounded-operator norm; differentiating on bounded time intervals gives $T_u f - f = \int_0^u T_v A f \, dv$. The fresh-future construction above therefore gives, conditionally on the observed past at s,

$$\mathbb{E}[f(X_t) - f(X_s) \mid \mathcal{F}_s] = (T_{t-s} f - f)(X_s) = \int_0^{t-s} T_v A f(X_s) \, dv.$$

Boundedness of Af permits conditional Fubini; the same expression is the conditional mean of $\int_s^t Af(X_r) dr$. Their difference has zero conditional mean. All observable and drift terms are integrable, the latter bounded by $2t\|f\|_\infty$, and the actual finite-jump paths make the drift adapted. This proves the martingale claim and agrees with the mapped native Dynkin theorem.

Finally every sum $\sum_{r=1}^3 u_r \xi^r$ has mark mean zero, whereas the constant one has mean one. No choice of pre-jump coefficients can equate them for every fresh mark. The count increases by exactly one at each arrival, and its continuous compensator has no jumps, so its compensated jump has this same missing constant component. □

The last obstruction is the explicit source statement about centered mark channels. It is not a proof that no representation exists after adding a clock channel, nor does the displayed decomposition establish a full wall-clock predictable representation theorem for arbitrary terminal functions of holdings and marks. We assert the combined martingale just proved; separate operator-domain and isometry claims for its four continuous-time components are not imported from the mark-index Hilbert space. Likewise no isometry is transferred to the signed heat target or to a nonexistent ordinary weak limit.

10 A genuine compact-frequency analytic domain

The failure of weak convergence does not prevent convergence on a proved class of unbounded complex tests. Fix $R \geq 0$ and $\varphi \in L^1([-R, R]; \mathbb{C})$, and put

$$f_\varphi(z) = \int_{-R}^{R} \varphi(\eta) e^{i\eta z} \, d\eta.$$

Lebesgue measure is used here, with no 2π normalization. At $R = 0$ this domain contains only the zero realization. For all $z \in \mathbb{C}$,

$$|f_\varphi(z)| \leq \|\varphi\|_1 e^{R|z|}, \qquad f_\varphi^{(k)}(z) = \int_{-R}^{R} (i\eta)^k \varphi(\eta) e^{i\eta z} \, d\eta. \tag{26}$$

Indeed $|e^{i\eta z}| = e^{-\eta \operatorname{Im} z} \leq e^{R|z|}$; on $|z - z_0| \leq 1$ each differentiated integrand is dominated by $R^k |\varphi(\eta)| e^{R(|z_0|+1)}$. Dominated differentiation proves the formula inductively and proves that f_φ is entire. Its real restriction is bounded, whereas its complex extension can grow.

Let $C_a(t)$ denote the actual unit-rate marked Poisson process of the preceding sections. The finite-rate compound law of $C_a(r)$ is $\mu_{r,a}$. Conditional on $N_r = j$, its norm is at most $2|a|j$, hence for every $B \geq 0$,

$$\mathbb{E} e^{B|C_a(r)|} \leq \exp\{r(e^{2B|a|} - 1)\} < \infty. \tag{27}$$

Thus f_φ, every derivative, and every shifted function $f_\varphi(z+\cdot)$ are genuinely integrable against each finite law. More precisely, the absolute double integral of the defining frequency integrand is at most

$$\|\varphi\|_1 e^{R|z|}\mathbb{E}e^{R|C_a(r)|} < \infty.$$

This is the Fubini justification; it is not inferred from cancellation in a holomorphic transform. With $\psi_a(\eta) = F(a\eta) - 1$, it gives

$$\mathbb{E}f_\varphi(z + C_a(r)) = \int_{-R}^{R} \varphi(\eta)e^{i\eta z}e^{r\psi_a(\eta)}\,d\eta. \tag{28}$$

For a real tilt κ, the density defining the finite complex law has the modulus bound

$$e^{-r\psi_a(\kappa)}e^{|\kappa||w|}.$$

Equation (27) with $B = R + |\kappa|$ therefore also proves variation-integrability before Fubini, and yields

$$\int f_\varphi(w)\,d\nu_{r,a,\kappa}(w) = \int_{-R}^{R} \varphi(\eta)e^{r(\psi_a(\eta+\kappa)-\psi_a(\kappa))}\,d\eta. \tag{29}$$

Here $d\nu_{r,a,\kappa} = e^{-r\psi_a(\kappa)}e^{i\kappa w}d\mu_{r,a}$; the notation includes exactly the rate-scaled tilts used earlier.

Theorem 10.1. *For $T \geq 0$, $z \in \mathbb{C}$ and every fixed integer $k \geq 0$,*

$$\mathbb{E}f_\varphi^{(k)}(z + C_{b_n}(nT)) \longrightarrow \int_{-R}^{R} (i\eta)^k\varphi(\eta)e^{i\eta z - T\eta^4}\,d\eta, \tag{30}$$

$$\int f_\varphi(w)\,d\nu_{nT,b_n,\kappa}(w) \longrightarrow \int_{-R}^{R} \varphi(\eta)e^{-T((\eta+\kappa)^4-\kappa^4)}\,d\eta. \tag{31}$$

Proof. The expansion $F(x) = 1 - x^4/6 + O(x^8)$ implies $F(x) \leq 1$ on a sufficiently small real neighborhood of zero. Since $b_n \to 0$, for every fixed frequency radius S and all sufficiently large n,

$$0 < e^{nT\psi_{b_n}(\eta)} \leq 1 \quad (|\eta| \leq S).$$

The pointwise limit is $e^{-T\eta^4}$. In (28), with $(i\eta)^k\varphi$ in place of φ, an eventual integrable majorant is $R^k|\varphi(\eta)|e^{R|z|}$. Dominated convergence proves the first limit, including $k = 0$ with its evident bound. For the tilt write its multiplier as

$$\frac{e^{nT\psi_{b_n}(\eta+\kappa)}}{e^{nT\psi_{b_n}(\kappa)}}.$$

The numerator has modulus at most one uniformly for $|\eta| \leq R$ eventually, by choosing $S = R + |\kappa|$. The denominator converges to $e^{-T\kappa^4} > 0$, so its reciprocal is bounded independently of n eventually. This supplies $C|\varphi|$ as an integrable frequency majorant. The pointwise ratio limit proves (31). No uniform bound on the variation of $\nu_{nT,b_n,\kappa}$ has been used; that variation diverges when $T > 0$ and $\kappa \neq 0$. □

The limit is a classical solution on this fixed spectral domain. Define

$$u(t,z) = \int_{-R}^{R} \varphi(\eta)e^{i\eta z - t\eta^4}\,d\eta \quad (t \in \mathbb{R}, z \in \mathbb{C}).$$

On bounded time and complex-coordinate sets, all time and space derivatives have integrable bounds by constants times $|\varphi|$. Consequently $u(0,z) = f_\varphi(z)$ and

$$\partial_t u(t,z) = -\partial_z^4 u(t,z).$$

Negative times are allowed here because the frequencies are bounded; this statement is not backward solvability on arbitrary initial data. In the library's 2π Fourier convention, the multiplier $e^{-T\eta^4}$ is the higher-heat multiplier at frequency $\eta/(2\pi)$.

An entire real-Schwartz test outside the finite-clock domain

Entire extension and rapid decay on the real line are insufficient for the preceding genuine process integrals. Take

$$g(z) = e^{-z^2}.$$

Its real restriction is Schwartz: every derivative is a polynomial times e^{-x^2}, and every polynomial multiple is bounded and decays at infinity. Yet, for every $r > 0$ and $a \neq 0$,

$$\int_{\mathbb{C}} |g(w)|\, d\mu_{r,a}(w) = +\infty. \tag{32}$$

Thus even a finite compound-root process time has no ordinary complex expectation of this test.

Here is the complete rare-path proof. Conditional on exactly $2m$ marks, prescribe the first m roots to be $1+i$ and the next m to be $-1+i$. This particular word has probability 4^{-2m} and endpoint $2mai$. At that endpoint

$$|g(2mai)| = e^{4a^2m^2}.$$

The nonnegative integral over the disjoint count events therefore dominates the sum, over $m \geq 0$, of

$$u_m = e^{-r}\frac{(r/4)^{2m}}{(2m)!}e^{4a^2m^2} > 0.$$

For fixed nonzero a and positive r its consecutive ratio is

$$\frac{u_{m+1}}{u_m} = \frac{(r/4)^2 e^{4a^2(2m+1)}}{(2m+2)(2m+1)} \longrightarrow +\infty.$$

For an elementary justification of the last divergence, use $e^x \geq x^3/6$ for $x \geq 0$: the resulting lower bound grows linearly in m. In particular $u_{m+1} \geq 2u_m$ eventually, so the terms do not tend to zero and their sum diverges. Tonelli for the nonnegative count mixture proves (32). The original process has exactly this law, so its test is nonintegrable as well. The exclusions are sharp for the degenerate cases $r = 0$ or $a = 0$, where the process value is identically zero and g equals one there.

For a deterministic finite walk every test value is integrable, since its sample space is finite. Even there, absolute control under the heat scaling can fail spectacularly. At $n = 2m$ and amplitude $a_{2m}(T)$, the same word gives, for fixed $T > 0$,

$$\mathbb{E}|g(S_{2m,a_{2m}(T)})| \geq \exp\{4\sqrt{3T}\, m^{3/2} - 2m\log 4\} \longrightarrow +\infty.$$

This scaled specialization is an elementary human deduction from the mapped finite-word lower bound, not an additional named formal limit. It concerns absolute expectation; it does not by itself determine a limit or divergence of the complex expectations with cancellation.

In particular, for every $n \geq 1$ and $T > 0$, the Poissonized test $g(C_{b_n}(nT))$ is already nonintegrable. No exchange of a full Gaussian frequency integral with these laws is justified by the earlier compact-frequency theorem. This is a finite-law integrability boundary, stronger than merely lacking a uniform bound in the approximation index. The spectral and entire-function hypotheses of a claimed representation must be checked before using its expectation notation. The restricted Fourier-data framework of [1] is not refuted by this example.

Source correspondence: `ComplexRootGaussianDomain.vertical_terminal`, `vertical_gaussian_norm`, `walk_norm_lower`, `not_integrable_compound`, `not_integrable_process`; `PoissonSuperexponentialObstruction.not_summable_subsequence`.

11 Actual conditional transitions and backward martingales

Fix the finite process parameter $a \in \mathbb{R}$ and write $\mathcal{H}_s^a = \sigma(C_a(v) : 0 \leq v \leq s)$ for its natural past. The renewal construction above supplies an increment independent of this entire past, with law $\mu_{t-s,a}$, for $0 \leq s \leq t$. Define the frequency propagator

$$(P_r^a \varphi)(\eta) = \varphi(\eta) e^{r\psi_a(\eta)}, \quad r \geq 0.$$

For fixed a, r, its multiplier is bounded on $[-R, R]$, so it preserves $L^1([-R, R])$. Direct multiplication gives $P_0^a = I$ and $P_r^a P_q^a = P_{r+q}^a$. It need not be a contraction on that domain at arbitrary finite a.

Theorem 11.1. *For $0 \leq s \leq t$,*

$$\mathbb{E}[f_\varphi(C_a(t)) \mid \mathcal{H}_s^a] = f_{P_{t-s}^a \varphi}(C_a(s)) \quad \textit{almost surely.}$$

For every fixed horizon $H \geq 0$, the process

$$M_t = f_{P^a_{(H-t)_+}\varphi}(C_a(\min(t, H))) \quad (t \geq 0)$$

is an integrable complex martingale in this same natural filtration.

Proof. The function $(p, w) \mapsto f_\varphi(p(s) + w)$ is measurable on the product of the coordinate past space and $\mathbb{C}$. Its value at the actual past and increment is $f_\varphi(C_a(t))$, which is integrable by (26)–(27). Independence therefore gives its conditional expectation by integrating only the increment. One may prove this conditional-integration rule first on product indicators, then on simple functions and finally on integrable functions by truncation and L^1 approximation. Formula (28) identifies this increment integral for every starting coordinate and proves the first assertion. This argument handles an unbounded test; bounded-continuous transition formulas alone would not justify it.

For $t \leq H$, the first assertion says exactly $M_t = \mathbb{E}[f_\varphi(C_a(H)) \mid \mathcal{H}_t^a]$. For $t \geq H$, $P_0^a = I$ and the terminal variable is already measurable in $\mathcal{H}_t^a$, so the same identity holds. The displayed version is adapted because the coordinate at $\min(t, H)$ is past-measurable and each propagated extension is continuous. Each time is integrable by the same finite-law bound. The tower property proves the martingale identity for every pair of times. □

These martingales concern each actual finite positive compound-root process. No limiting path law, limiting filtration or limiting martingale representation is supplied by the expectation limits. The compact-frequency Fourier mechanism has a direct antecedent: Bonaccorsi–Mazzucchi [1],

Hypothesis 10 and Theorem 11, already represent admissible Fourier-measure data and the corresponding higher-order equation. Compactly supported L^1 densities meet those spectral integrability conditions. Our Poisson clock, finite conditional identities and fully specified domination reconstruct the frozen library's version; they do not establish a new analytic representation principle or priority over that work.

12 Classical finite-state context

Discrete Fourier orthogonality, finite-state martingale representation and its Hilbert completion have classical antecedents. Franz–Hamdi [3], Proposition 2.3 and Propositions 7.1–7.5, already give predictable isometry and full square-integrable reconstruction in a more general real obtuse-walk setting. Our four-mark space can equally be expressed with real coordinates $\mathrm{Re}(\xi)/a$, $\mathrm{Im}(\xi)/a$ and their product: these are three centered orthonormal sign functions on the four points. They span the same centered complex function space as the three powers above. This change of basis is an elementary deduction, not a new normal-martingale representation principle. Attal–Deschamps–Pellegrini [4], Sections 2.2–2.4, develop complex finite-state orthonormal-coordinate structure much more broadly. Their tensor and continuous-limit theorems are not needed for the finite character calculations here.

The contribution of these added sections is a complete, domain-specific reconstruction on the actual original source, with exact normalizations, finite-energy completion and an explicit mark/clock distinction. The bounded primary comparison establishes strong classical antecedents; it does not certify novelty of the assembled exposition. No new Clark formula, cyclic Fourier transform or compound-Poisson calculus is claimed.

13 Formal correspondence and limits of the conclusions

All formal statements used here are frozen at verified commit `bab688bb6140cd6316c80680095a914ea0d346db`. The accompanying machine-readable theorem map records exact declaration names, locations and source hashes. Human proofs reorganize the finite sign computations and the weak-limit contradiction; they introduce no new Lean theorem or hidden probabilistic axiom.

The formal continuous walk paths are defined on all of $\mathbb{R}$, are constant outside $[0,1]$, and carry the compact-open topology. Restriction to $[0,1]$ gives exactly the polygonal path above. Corollary 4.1 proves its uniform-topology version directly from the verified endpoint law; the general continuous-coordinate implication is a classical deduction, not an additional named formal theorem.

The additional compact-frequency correspondences cover the exponential growth domain, entire differentiation, absolute product integrability, actual finite conditional transitions, all derivative expectation limits and the classical fourth-order equation. Their proofs above retain a fixed spectral radius. The source-calculus correspondences additionally include `CyclicMarkRepresentation`, `RootMarkInfiniteIto`, `RootMarkInfiniteClarkOperator`, `MarkedChannelIto` and `CyclicClockChannelObstruction`. Their exact original probability spaces and domains remain separate from the signed target. The central approximation modules are `ComplexRootHeatTransformLimit`, `ComplexRootPoissonNoWeakLimit`, `ComplexRootTiltedLaw` and `ComplexTiltVariationDivergence`. The theorem map also identifies the exact finite transform, moment, path, phase-cancellation and variation declarations underlying each proof.

The Poisson compact-escape result uses rate n and amplitude $a_n(T)$; the tilted limit uses rate nT and amplitude b_n. These normalizations give the same analytic heat symbol, but are not the same finite law. They have been kept separate throughout the statements and proofs.
The complete baseline build and axiom receipts are hash-checked; no full Lean rebuild is run for these publication-only edits. No continuum probability process with biharmonic transitions, unrestricted analytic calculus or stochastic calculus for an ordinary weak limit is proved. Attribution, referee review and comprehensive novelty assessment remain open.

References

[1] S. Bonaccorsi and S. Mazzucchi, *High order heat-type equations and random walks on the complex plane*, arXiv:1402.6140 (2014), arxiv.org/abs/1402.6140. Primary comparison: Theorem 2, Remark 1, the discussion before Theorem 7, and Hypothesis 10 and Theorem 11.

[2] S. Bonaccorsi, C. Calcaterra and S. Mazzucchi, *An Itô calculus for a class of limit processes arising from random walks on the complex plane*, arXiv:1603.05437 (2016), arxiv.org/abs/1603.05437.

[3] U. Franz and T. Hamdi, *Stochastic analysis for obtuse random walks*, primary preprint, arXiv:1212.2324, Proposition 2.3 and Propositions 7.1–7.5.

[4] S. Attal, J. Deschamps and C. Pellegrini, *Complex obtuse random walks and their continuous-time limits*, primary preprint, arXiv:1312.3731, Sections 2.2–2.4.